Chapter 3

3. Principal component analysis.

- 3.1. Motivation, definition and computation.
- 3.2. Properties and interpretation.
- 3.3. Empirical version.
- 3.4. National record data.
- 3.5. NO2 data.

Motivation, definition and calculation

We analyse a p-random vector X. Potentially p is large and we want to extract relevant information of $X \Longrightarrow$ *data reduction*.

The information can be captured in a new vector of lower dimension q.

Preferably q much smaller than p if p is large.

Motivation, definition and calculation

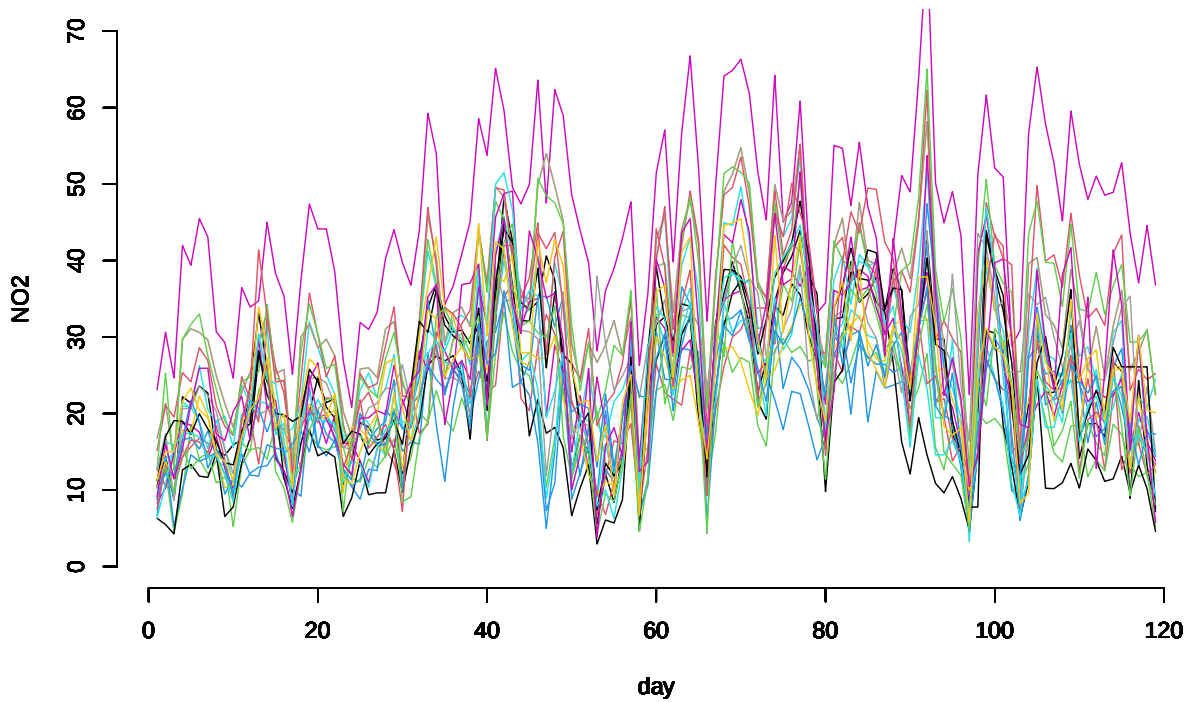
The procedure should work without loosing much of the "information" contained in the original vector X.

By the dimension reduction we hope to

- get better/easier interpretation (think of plotting) of the data.
- make data easier to process and statistically analyse.

Motivation, definition and calculation

Example. NO2 concentration in 23 measuring sites across Styria from Nov 1 2019, to Feb 28 2020 ($n = 119$).



Motivation, definition and calculation

Obviously, the data are highly correlated.

Visual inspection here is doable, since we have a time series structure and we have 23 similar curves. In general visualisation is difficult.

Motivation, definition and calculation

Note that both, the data matrix X and also X' can be analysed.

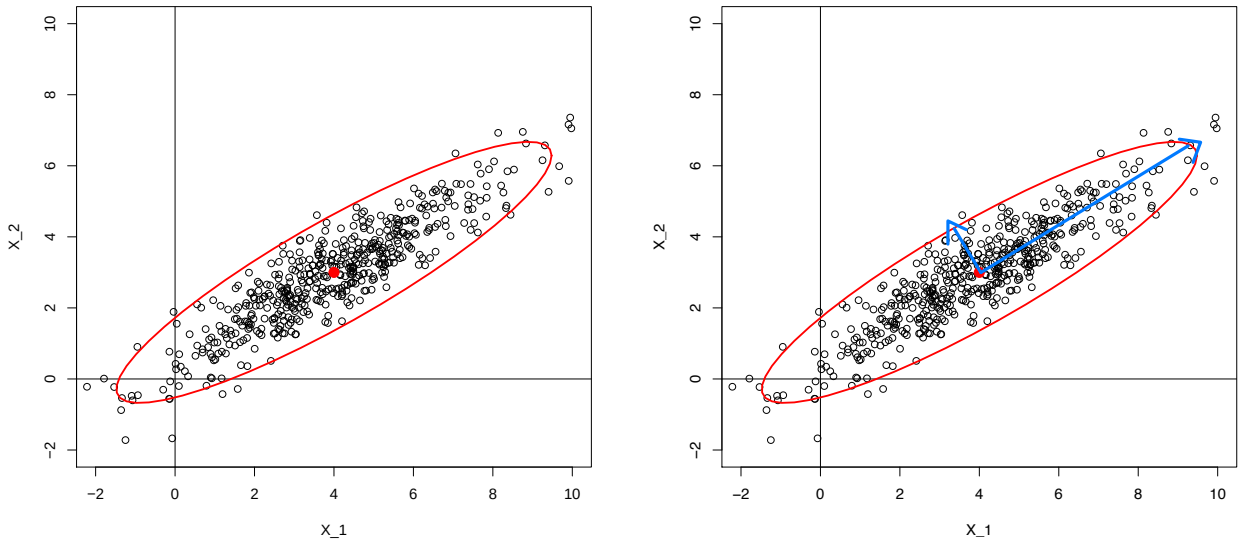
X: we have 119 measurments with 23 components. The components refer to measuring sites. (Temporal data).

X': we have 23 measurements with 119 components. The components refer to time. (Spatial data).

Question: can we find a good representation in lower dimensions?
We will investigate these data at the end of the section.

Motivation, definition and calculation

To get the basic idea behind PCA look at the bivariate normal data from Chapter 1.



We see that components X_1 and X_2 are obviously correlated and hence contain mutual information. By rotating the data we can make them vary along the main axes, and hence make the components uncorrelated.

Motivation, definition and calculation

In mathematical terms this rotation of the vector X we will be achieved by a linear mapping H

$$X \mapsto HX = Y, \quad Y \in \mathbb{R}^p.$$

We will retain the first q components of Y, which are the components that relate to the main axes of variation.

Motivation, definition and calculation

The rotation matrix to use is suggested by the spectral theorem.

Recall that $\Sigma = \mathrm{Var}(X) = O\Lambda O'$. If we rotate with O' we get

$$\mathrm{Var}(O'X) = O'O\lambda O'O = \Lambda,$$

and hence the variance of $Y = O'X$ has diagonal form.

Note: It is common to first center the data, before we rotate. This leads to the following definition:

Motivation, definition and calculation

Definition

Let X be a p-vector with mean μ and covariance matrix Σ. Let $\Sigma = O\Lambda O'$ be the spectral representation of Σ. Then the transformation

$$X \mapsto Y = O'(X - \mu) = \begin{pmatrix} e_1' \\ \vdots \\ e_p' \end{pmatrix} (X - \mu) = \begin{pmatrix} e_1'(X - \mu) \\ \vdots \\ e_p'(X - \mu) \end{pmatrix}$$

is called principal component transformation.

We call Y_i the i-th principal component (score) of X and e_i the i-th principal component loading vector of X.

Properties and interpretation

Proposition

Let Σ be the covariance matrix of some p-vector X and assume that $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$ are its eigenvalues and that $O = (e_1, \ldots, e_p)$ is the matrix of the corresponding eigenvectors. Let Y be the principal components of X. Then

$$(i) \quad \mathrm{Var}(Y) = \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_p),$$

$$(ii) \quad \mathrm{tr}(\Lambda) = \mathrm{tr}(\Sigma) = E\|X - \mu\|^2,$$

$$(iii) \quad |\Lambda| = |\Sigma|.$$

Properties and interpretation

Proof. Using the spectral theorem and $O'O = I_p$ we have

$$\mathrm{Var}(Y) = \mathrm{Var}(O'X) = O'\Sigma O = O'O\Lambda O'O = \Lambda,$$

showing (i). Next,

$$\mathrm{tr}(\Sigma) = \mathrm{tr}(O\Lambda O') = \mathrm{tr}(O'O\Lambda) = \mathrm{tr}(\Lambda),$$

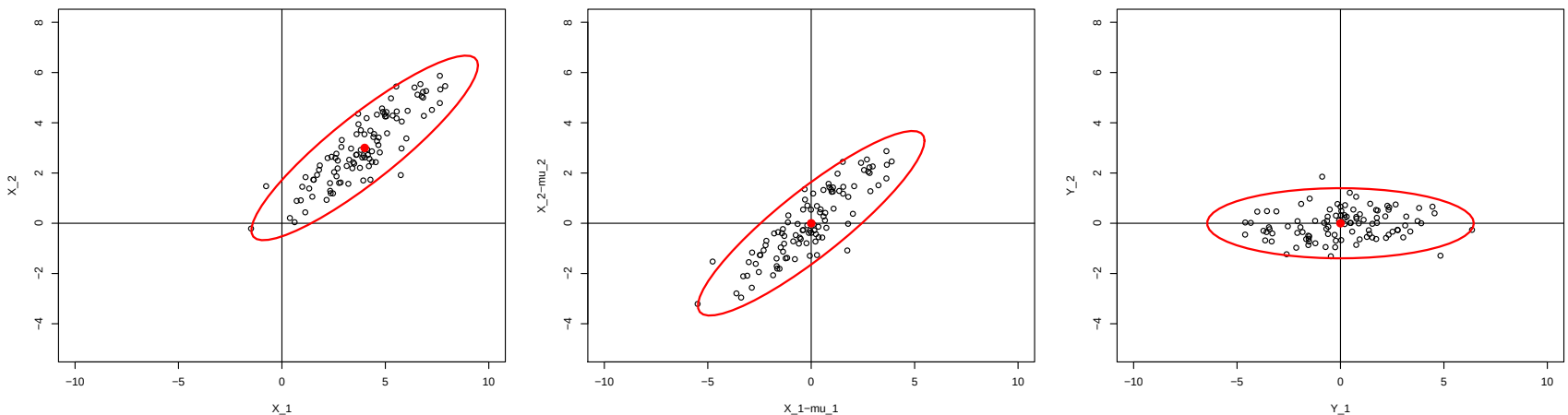
and

$$|\Sigma| = |O\Lambda O'| = |O||\Lambda||O'| = |\Lambda|,$$

showing (ii) and (iii). □

Properties and interpretation

The figure below shows the principal component transformation applied to 100 data from a $\mathcal{N}_p(\mu, \Sigma)$ distribution with $\mu = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\Sigma = \begin{pmatrix} 5 & 3 \\ 3 & 2.25 \end{pmatrix}$.



Properties and interpretation

Alternatively PCA can be introduced using the following algorithm:

(1) Start with a p-vector h such that

$$\mathrm{Var}(Y_1) = \mathrm{Var}(h'X)$$

is maximal with respect to the constraint $\|h\| = 1$. (Without constraint this doesn't make sense!)

$$h_1 = \mathrm{argmax} \left\{ \mathrm{Var}(h'X) : h \in \mathbb{R}^p, \|h\| = 1 \right\}.$$

In other words: $Y_1 h$ is the projection of X onto the 1-dimensional axis, on which we have the highest possible variability.

Motivation, definition and calculation

(2) Define

$$h_2 = \operatorname{argmax} \left\{ \operatorname{Var}(h'X) : h \in \mathbb{R}^p, \, \|h\| = 1, h \perp h_1 \right\}.$$

In other words: $Y_2 h_2$ is the projection of X onto the 1-dimensional axis, which is orthogonal to h_1 and on which we have the highest possible variability.

(3) Define

$$h_3 = \operatorname{argmax} \left\{ \operatorname{Var}(h'X) : h \in \mathbb{R}^p, \, \|h\| = 1, h \perp \operatorname{sp}\{h_1, h_2\} \right\}.$$

(4) Continue in the same way to obtain $h_4, \ldots, h_p$.

The solution to this exercise turns out to be $e_1, \ldots, e_p$.

Properties and interpretation

So far we didn't discard any variables of the transformed data. We can still fully recover X from Y:

$$X = OY + \mu. \tag{1}$$

We wish now to replace the vector $Y = (Y_1, \ldots, Y_p)$ by a lower dimensional vector $Y^{(q)} = (Y_1, \ldots, Y_q)$, $q \leq p$.

Since $O = (e_1, \ldots, e_p)$ we get from (1) that

$$X - \mu = Y_1 e_1 + Y_2 e_2 + \cdots + Y_p e_p.$$

If we only keep $Y_1, \ldots, Y_q$ we could use as an approximation

$$X - \mu \approx Y_1 e_1 + Y_2 e_2 + \cdots + Y_q e_q.$$

Properties and interpretation

The following proposition states, that the approximation

$$X - \mu \approx Y_1 e_1 + Y_2 e_2 + \cdots + Y_q e_q.$$

is in some sense the best possible in terms of q basis vectors.

Proposition

Let $B = (b_1, \ldots, b_p)$ be any orthonormal basis of $\mathbb{R}^p$. Then

$$X - \mu = W_1 b_1 + W_2 b_2 + \cdots + W_p b_p,$$

with $W_k = b_k'(X - \mu)$ and for all $1 \leq q \leq p$

$$E \left\| (X - \mu) - \left(\sum_{i=1}^{q} W_i b_i \right) \right\|^2 \geq E \left\| (X - \mu) - \left(\sum_{i=1}^{q} Y_i e_i \right) \right\|^2 .$$

Properties and interpretation

We note that

$$E\left\|(X-\mu)-\left(\sum_{i=1}^{q} Y_i e_i\right)\right\|^2 = \lambda_{q+1}+\ldots+\lambda_p.$$

The ratio

$$\frac{\lambda_1+\ldots+\lambda_q}{\lambda_1+\ldots+\lambda_p} = \frac{\lambda_1+\ldots+\lambda_q}{\mathrm{tr}(\Sigma)}, \tag{2}$$

is a measure for the amount of information explained by the first q principal components. One usually says that (2) is the proportion of variance explained by the first q PCs.

Properties and interpretation

Example. Suppose $X = (X_1, X_2, X_3)'$ with zero mean and covariance matrix

$$\Sigma = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Then the eigenvalues are given by

$$(\lambda_1, \lambda_2, \lambda_3) = (5.83, 2, 0.17)$$

and the eigenvectors are

$$e_1 = \begin{pmatrix} 0.383 \\ -0.924 \\ 0 \end{pmatrix} \quad e_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad e_3 = \begin{pmatrix} 0.923 \\ 0.383 \\ 0 \end{pmatrix}.$$

Properties and interpretation

This gives

$$Y_1 = 0.383X_1 - 0.923X_2$$
$$Y_2 = X_3$$
$$Y_3 = 0.924X_1 + 0.383X_2$$

The coordinates of the eigenvectors tell us how much weight is attributed to each X_i!

In this example the proportion of the total variance explained by the first PC is $\lambda_1/\mathrm{tr}(\Sigma) = 5.83/8 = 0.73$, the one explained by the first 2 PCs is $(\lambda_1 + \lambda_2)/\mathrm{tr}(\Sigma) = (5.83 + 2)/8 = 0.98$.

Properties and interpretation

An unfavorable property of PCA is that it is *not scale-invariant*. This means that if A is a scaling matrix and $\mathcal{P}$ denotes the principal component transformation, then

$$\mathcal{P}(AX) \neq A\mathcal{P}(X).$$

Example. Let $EX = 0$ and $\mathrm{Var}(X) = \begin{pmatrix} 1 & 4 \\ 4 & 100 \end{pmatrix}$. The eigenvalues are $\lambda_1 = 100.16$ and $\lambda_2 = 0.84$ and the eigenvectors are

$$e_1 = \begin{pmatrix} 0.04 \\ 0.999 \end{pmatrix} \quad \text{and} \quad e_2 = \begin{pmatrix} 0.999 \\ -0.04 \end{pmatrix}$$

We have $\lambda_1/\mathrm{tr}(\Sigma) = 0.992$.

The large variance of X_2 completely dominates the first principal component derived from Σ!

Properties and interpretation

We use the same setup, but we first standardize X by dividing the components by its standard deviation. This means we use the correlation matrix

$$P = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1 \end{pmatrix}$$

to determine the principal components. The eigenvalues now are $\lambda_1 = 1.4$ and $\lambda_2 = 0.6$ and the eigenvectors are

$$e_1 = \begin{pmatrix} 0.707 \\ 0.707 \end{pmatrix} \quad \text{and} \quad e_2 = \begin{pmatrix} 0.707 \\ -0.707 \end{pmatrix}$$

We have $\lambda_1/\mathrm{tr}(P) = 0.7$.

We attribute the same importance to X_1 and X_2 in the calculation of Y_1.

Properties and interpretation

This example shows that PCA should be taken with care.

PCA can be best used if all variables are on a comparable scale.

Variables should be properly standardized if they are measured on scales with widely different ranges.

Sample PCA

Of course, in practice we don't know μ and Σ and thus we have to base our analysis on their estimates

$$\bar{X} \quad \text{and} \quad S,$$

which both are computed from the sample $X_1, \ldots, X_n$.

Let $\hat{O} = (\hat{e}_1, \ldots, \hat{e}_p)$ be the standardized eigenvectors of S belonging to the eigenvalues $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \ldots \geq \hat{\lambda}_p$. The empirical principal component transformation is given by

$$X_k \mapsto \hat{Y}_k = \hat{O}'(X_k - \bar{X}), \quad 1 \leq k \leq n.$$

Letting $\hat{Y}_k = (\hat{Y}_{k,1}, \ldots, \hat{Y}_{k,p})'$, the variable $\hat{Y}_{k,i}$ is called the *i*-th empirical principal component (score) of X_k.

Sample PCA

Proposition

Let S_X be the covariance matrix of the random sample $X_1, \ldots, X_n$. Consider the spectral decomposition $S = \hat{O}\hat{\Lambda}\hat{O}'$. Then

$$(i) \quad S_Y = \hat{\Lambda},$$

$$(ii) \quad \operatorname{tr}(\hat{\Lambda}) = \operatorname{tr}(S_X) = \frac{1}{n-1}\sum_{k=1}^{n} \|X_k - \bar{X}\|^2,$$

$$(iii) \quad |\hat{\Lambda}| = |S_X|.$$

Proof. *Exercise.* □

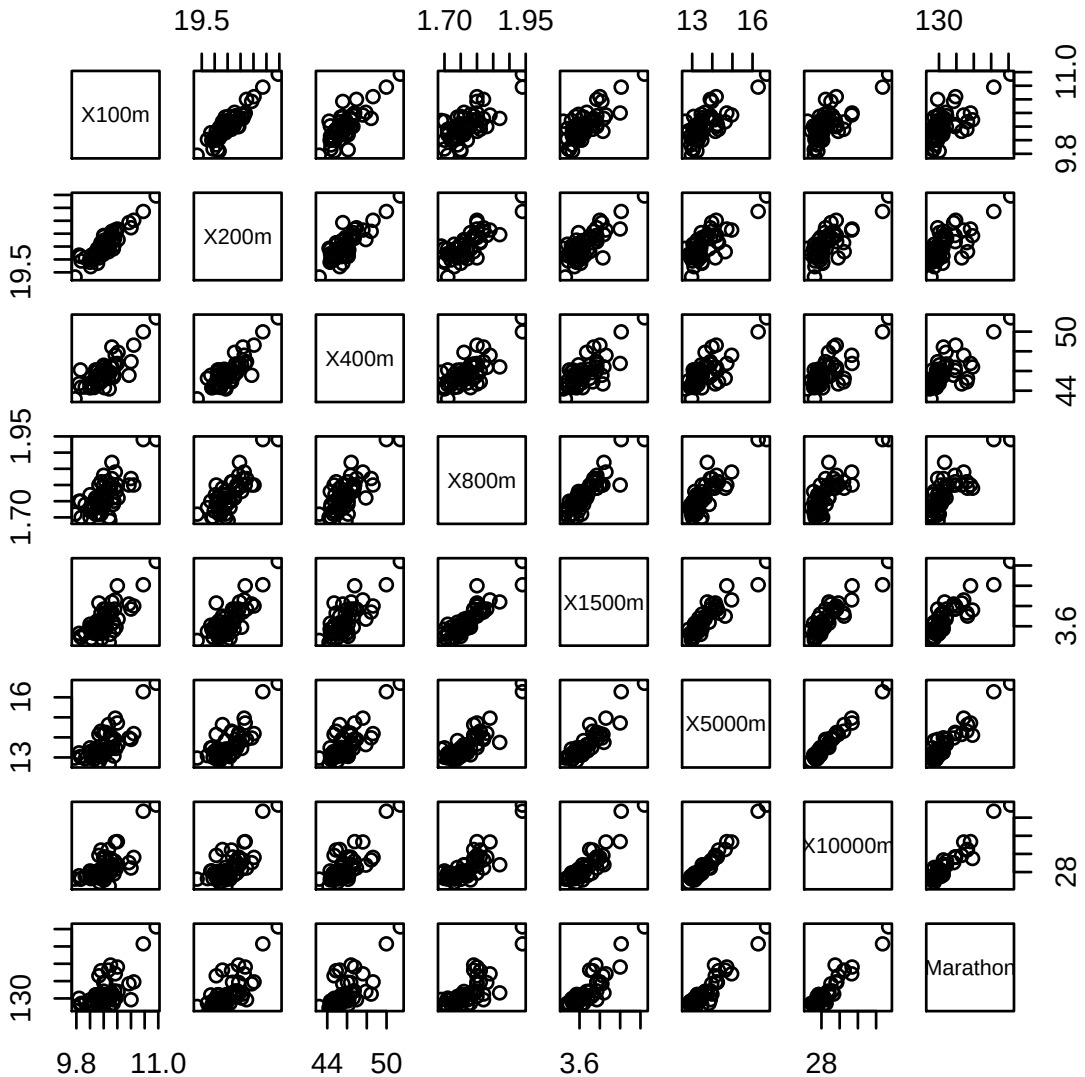
Track record data

We analyse the national track record data of 54 countries.

```
# read the track data
> track<-read.table("T8-6.dat",col.name=c("country","100m","200m",
              +  "400m","800m","1500m","5000m","10000m","Marathon"))
> head(track)
    country X100m X200m X400m X800m X1500m X5000m X10000m Marathon
1 Argentina 10.23 20.37 46.18  1.77   3.68  13.33   27.65   129.57
2 Australia  9.93 20.06 44.38  1.74   3.53  12.93   27.53   127.51
3   Austria 10.15 20.45 45.80  1.77   3.58  13.26   27.72   132.22
4   Belgium 10.14 20.19 45.02  1.73   3.57  12.83   26.87   127.20
5   Bermuda 10.27 20.30 45.26  1.79   3.70  14.64   30.49   146.37
6    Brazil 10.00 19.89 44.29  1.70   3.57  13.48   28.13   126.05
.....

> pairs(track[,2:9])
```

Track record data



Track record data

```
> x <-track[,2:9]
> pr <- prcomp(x)
> pr
Standard deviations (1, .., p=8):
[1] 9.19278216 1.06833766 0.47829361 0.28384152 ...

Rotation (n x k) = (8 x 8):
                  PC1         PC2          PC3 ...
X100m    -0.016521364 -0.09994775 -0.008058901 ...
X200m    -0.043603961 -0.25282207 -0.080820783 ...
X400m    -0.113581901 -0.91633442 -0.253458358 ...
X800m    -0.004643738 -0.01405243  0.012268779 ...
X1500m   -0.014583470 -0.03076566  0.037064086 ...
X5000m   -0.078593405 -0.11653932  0.376765236 ...
X10000m  -0.175189411 -0.20893763  0.872775053 ...
Marathon -0.973561706  0.16745845 -0.154755649 ...
```

Track record data

The summary shows, that the first PC explains almost all the variance.

```
# getting the amount of variance explained
> summary(pr)
Importance of components:
                          PC1     PC2     PC3     PC4 ...
Standard deviation     9.1928 1.06834 0.47829 0.28384 ...
Proportion of Variance 0.9829 0.01327 0.00266 0.00094 ...
Cumulative Proportion  0.9829 0.99615 0.99881 0.99975 ...
```

Track record data

- The PCs are not easy to interpret.
- The length of the track also the range changes drastically.
- It is not a surprise that the contribution of the marathon dominates the value of the first PC scores.

Track record data

We now perform a PCA with the standardized data.

```
> x <- scale(x)
> pr <- prcomp(x)
> pr
Standard deviations (1, .., p=8):
[1] 2.58907125 0.79900570 0.47699528 0.45370605 ...

Rotation (n x k) = (8 x 8):
                PC1         PC2          PC3         PC4
X100m    -0.3323877 -0.52939911 -0.343859303 -0.38074525 ...
X200m    -0.3460511 -0.47039050  0.003786104 -0.21702322 ...
X400m    -0.3391240 -0.34532929  0.067060507  0.85129980 ...
X800m    -0.3530134  0.08945523  0.782711152 -0.13427911 ...
X1500m   -0.3659849  0.15365241  0.244270040 -0.23302034 ...
X5000m   -0.3698204  0.29475985 -0.182863147  0.05462441 ...
X10000m  -0.3659489  0.33360619 -0.243980694  0.08706927 ...
Marathon -0.3542779  0.38656085 -0.334632969 -0.01812115 ...
```

Track record data

The summary shows, that the first 2 PCs explain 90% of the variance.

```
# getting the amount of variance explained
> summary(pr)
Importance of components:
                          PC1    PC2     PC3     PC4
Standard deviation     2.5891 0.7990 0.47700 0.45371 ...
Proportion of Variance 0.8379 0.0798 0.02844 0.02573 ...
Cumulative Proportion  0.8379 0.9177 0.94615 0.97188 ...
```

Track record data

Interpretation:

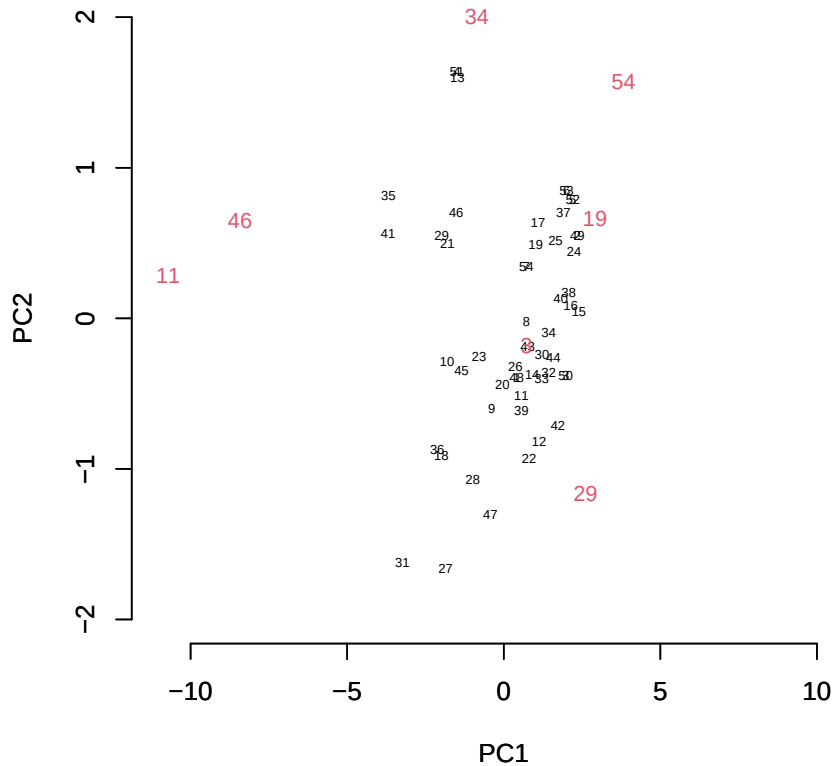
- The first PC is formed by an average over all disciplines. If the first PC is small/large, the nation is bad/good in all disciplines.
- The second PC contrasts the short distances from the long distances. A large second PC indicates, that a nation has strong records in short distance, but not so strong in long distance and vice versa.
- The 3rd PC contrasts middle length distances against short and long distances.

Track record data

With a biplot we can show the first two scores by country.

```
# define scores
> O <- as.matrix(pr$rotation)
> sco <- x%*%O
> plot(sco[,1],sco[,2],type="n",xlab="PC1",ylab="PC2",bty="n")
> text(sco[,1],sco[,2], labels=1:54, cex=0.5)
```

Track record data



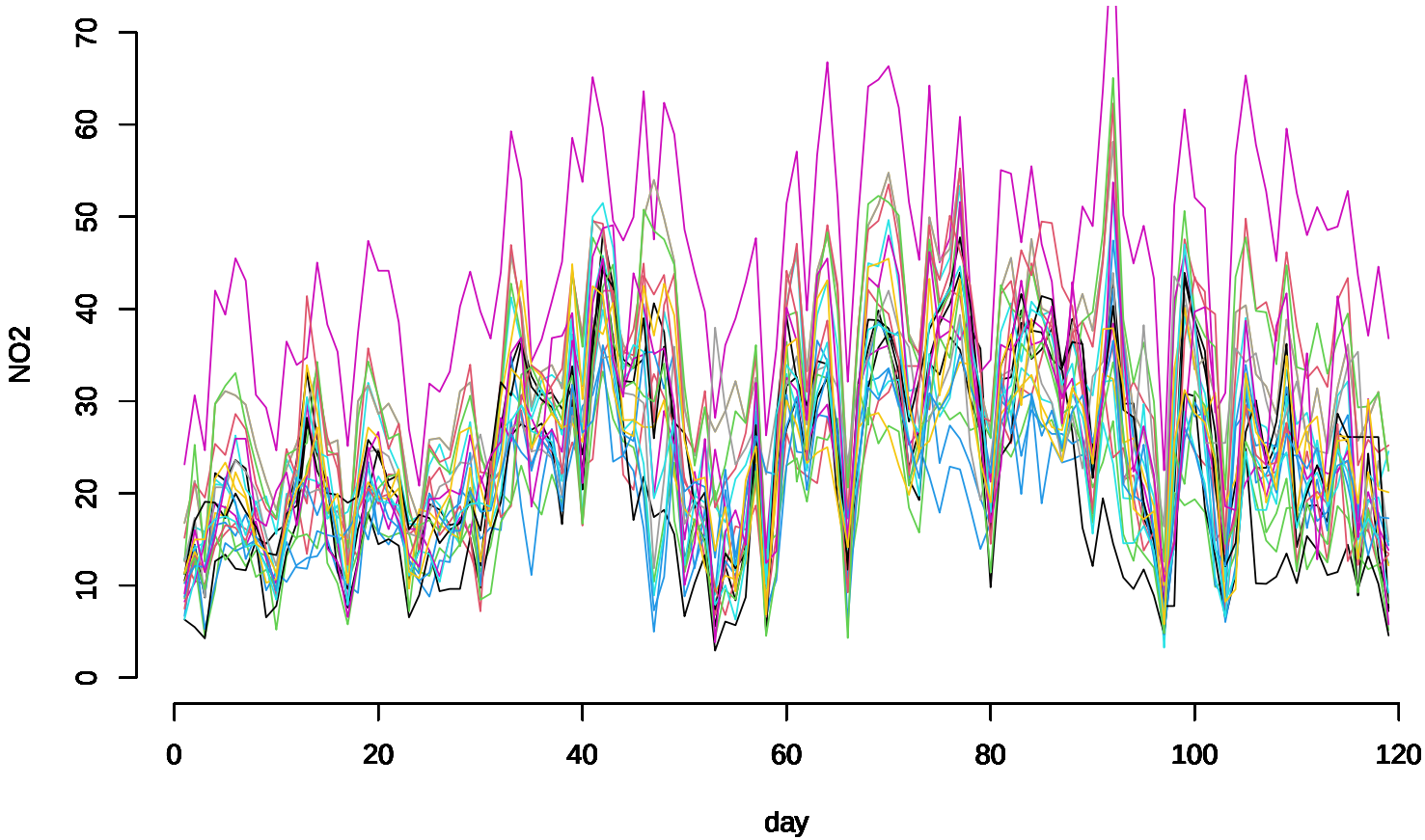
Some notable countries: 54=USA, 19=Great Britain, 11=Cook Islands, 46=Samoa, 29=Kenya, 34=Mauritius, 3=Austria

NO2 data

In our next example we study the NO2 data from Styria.

```
# read csv file
> no2 <- read.csv("NO2ST.csv", sep=";", dec=",", header=TRUE)
# remove the first column (date)
> no2 <- no2[,-1]
# this is the joint plot of the time series
> for(i in 1:23){
+   plot(no2[,i],col=i,ylim=c(0,70),type="l",ylab="NO2",bty="n",xlab="day")
+   par(new=TRUE)
+ }
```

NO2 data



In this example our components all are on a comparable scale.

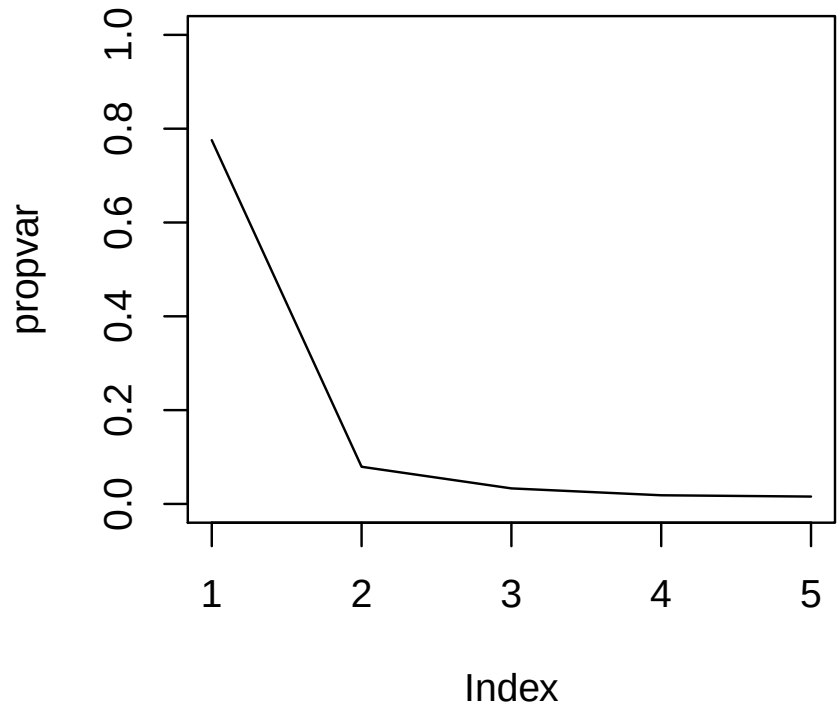
NO2 data

We produce a screeplot showing the proportion of variance explained by the PCs.

A rule of thumbs is to focus on the number of components up to the kink.

```
> pctemp <- prcomp(no2)
> lambdas <- pctemp$sdev[1:5]^2/sum(pctemp$sdev^2)
> plot(lambdas,ylim=c(0,1),type="l",ylab="propvar")
```

NO2 data



In this example, one or two components should be sufficient.

NO2 data

In order to better interpret the principal components we plot the empirical mean vector $\bar{X}$ along with $\bar{X} \pm \sqrt{\hat{\lambda}_i}\hat{e}_1 i$, $i = 1, 2$.

```
> mu <- as.vector(colMeans(no2))
> muplus <- mu+pctemp$rotation[,1]*pctemp$sdev[1]
> muminus <- mu-pctemp$rotation[,1]*pctemp$sdev[1]
>
> plot(mu,ylim=c(0,60),col=2,type="b",ylab="")
> text(muplus,label="+")
> text(muminus,label="-")
```

NO2 data

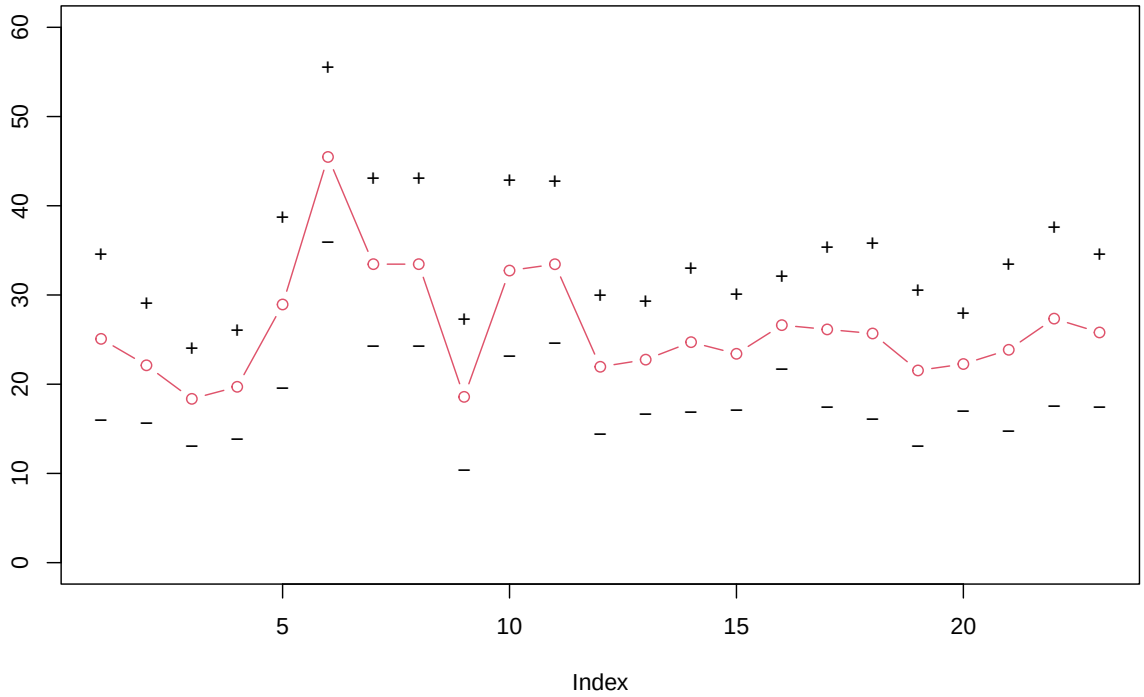


Figure 1: $\bar{X}$ and $\bar{X} + \sqrt{\hat{\lambda}_1}\hat{e}_1$ (+) and $\bar{X} - \sqrt{\hat{\lambda}_1}\hat{e}_1$ (-).

We conclude: the first PC refers to an overall level. Large score means, all stations have a higher load and vice versa.

NO2 data

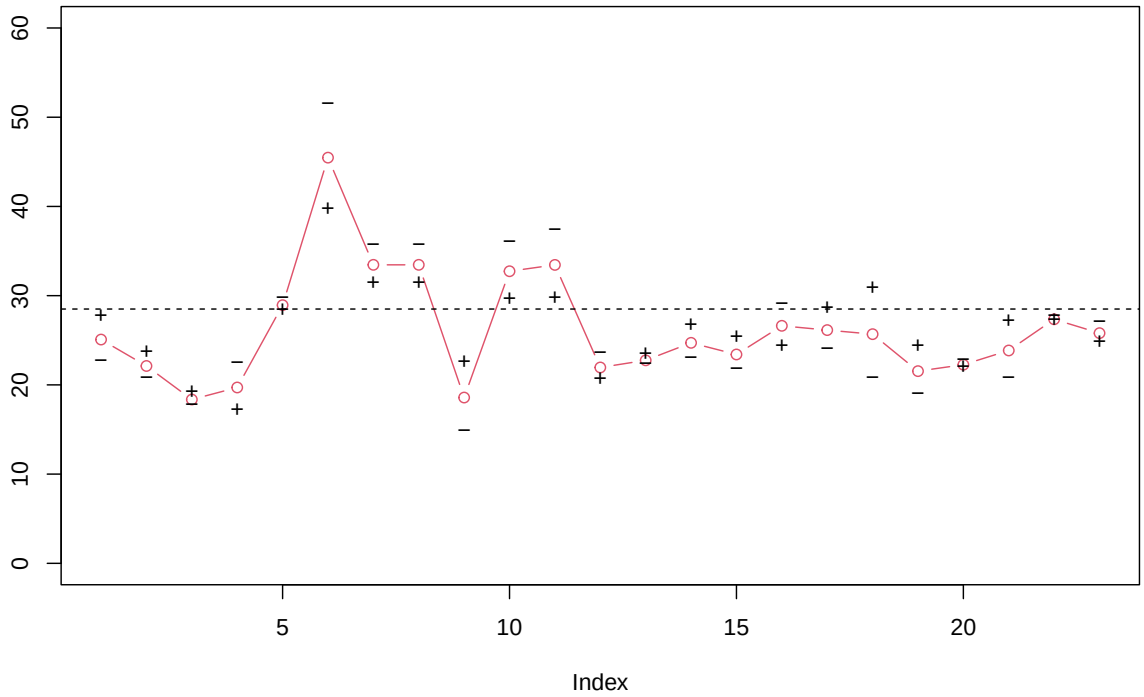


Figure 2: $\bar{X}$ and $\bar{X} + \sqrt{\hat{\lambda}_2}\hat{e}_2$ (+) and $\bar{X} - \sqrt{\hat{\lambda}_2}\hat{e}_2$ (-).

The second PC contrast stations with overall high load, to those with overall lower load.